

Companion LXXVIII: Schrödinger Bridges and Diffusion-Regularized Transport of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Schrödinger bridges provide a dynamic and stochastic generalization of entropic optimal transport, describing the most likely diffusion process connecting two probability distributions. Applied to persistence diagrams, Schrödinger bridges yield diffusion-regularized transport paths that interpolate between topological ensembles in a smooth and probabilistically meaningful way. Building on the Sinkhorn geometry developed in Companion LXXVII, this work introduces Schrödinger bridge dynamics for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the temporal evolution of the gravitational potential, inducing characteristic changes in diffusion flows, stochastic transport costs, and dynamic barycenters. The resulting framework provides a powerful tool for analyzing RDCC signatures in weak-lensing topology through the lens of stochastic optimal transport.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Classical Wasserstein geometry measures the optimal transport cost between diagrams, while entropic optimal transport introduces a regularization that yields smooth and computationally efficient Sinkhorn divergences.

Schrödinger bridges extend this framework by introducing a dynamic, stochastic interpretation: instead of computing a static transport plan, one seeks the most likely diffusion process that evolves one distribution of persistence diagrams into another. This dynamic viewpoint is particularly relevant for cosmology, where the underlying gravitational potential evolves over time.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation mechanism modifies the temporal structure of the gravitational field, producing characteristic signatures in diffusion-regularized transport of topological features.

2 Schrödinger Bridges

Given two probability distributions ρ_0 and ρ_1 on persistence diagrams, the Schrödinger bridge problem seeks the most likely diffusion process $(\rho_t)_{t \in [0,1]}$ connecting them.

The bridge solves

$$\min_{\rho_t, v_t} \int_0^1 \int \frac{1}{2} \|v_t(D)\|^2 \rho_t(D) dD dt + \varepsilon \text{KL}(\rho_t \| \rho_{\text{ref}}), \quad (1)$$

subject to the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad (2)$$

and boundary conditions

$$\rho_0, \quad \rho_1.$$

Here ε controls the diffusion strength.

3 Diffusion-Regularized Transport of Persistence Diagrams

For persistence diagrams, the diffusion process acts on birth-death coordinates:

$$dX_t = v_t(X_t) dt + \sqrt{2\varepsilon} dB_t, \quad (3)$$

where B_t is Brownian motion on the diagram space.

The resulting transport path (ρ_t) is smoother and more stable than classical Wasserstein geodesics, making it well-suited for noisy weak-lensing data.

4 RDCC Modification of Schrödinger Dynamics

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

This modifies the drift term in the Schrödinger bridge:

$$v_t^{\text{RDCC}}(D) = v_t(D) \left[1 - \chi_v \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

The diffusion-regularized transport cost becomes

$$\mathcal{S}^{\text{RDCC}} = \mathcal{S} \left[1 - \chi_S \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

5 Dynamic RDCC Signatures

RDCC induces:

- slower diffusion at small scales,
- enhanced drift at large scales,
- a characteristic turnover near ℓ_{rel} ,
- modified Schrödinger barycenters,
- altered stochastic geodesics in topological space,
- scale-dependent deformation of diffusion paths across redshift.

These signatures provide a dynamic and probabilistic probe of the RDCC gravitational field.

6 Conclusion

Schrödinger bridges provide a dynamic and stochastic generalization of entropic optimal transport, yielding diffusion-regularized transport paths between persistence diagrams. In the Relaxation-Driven Cyclic Cosmology, the relaxation mechanism modifies these diffusion flows in a scale-dependent manner, producing distinctive signatures in weak-lensing topology. The present Companion establishes the foundation for future studies of stochastic optimal transport, diffusion models, and dynamic geometric analysis in cosmological topology.

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